Groundwork: invoicing

Outside-in Radar from public data. No interviews and no private customer data.

0 of 10
riskiest Job guesses supported by public evidence (0 contradicted)

7 of 10
Job guesses have no public evidence in this harvest

NOT DONE
human label check

No money figure
every Q1–Q9 is S0, so synthetic validation fails

What this means

- **Decision: NO GO to build: run the cheapest market touch first.** The cheapest market touch is the T4 landing-page test (not run yet; B7): 3 headlines, EUR 49 vs EUR 149, about 200 founders or PMs.
- The 10 riskiest guesses about the customer's Job all have status **unknown**. J2, J3 and S2 have some supporting posts, below the bar of 3 distinct authors in 2 customer-voice families. The other 7 have no public evidence in this harvest.
- What the data does show: the problems 1,859 app reviewers describe, grouped into 12 themes, and which other options they name.
- All precision figures are against model labels: model-labelled (moonshotai/kimi-k3.1-pro), not human-verified; precision against model labels; cross-model agreement with z-ai/glm-5.3 89.7% (kappa 0.86); human check NOT DONE.
- **Simulation is exploratory and unvalidated.** It upgrades no node and feeds no estimate. Its section is at the end; every number there is labelled **SYNTHETIC (simulation estimate)**.

Scope: Radar (problems and alternatives); T2 failed, so no Job node has a status.

1. Problem radar

Pool: 3,284 posts (appstore 3,283, hackernews 1). The producer model called 3,135 of them first-hand users and found a problem in 1,868 (1,859 distinct authors). Themes come from k-means (k = 12, fixed in advance) on text embeddings; theme names are model labels (Kimi), not findings. Counts are voices in this pool, not market prevalence.

Theme	Voices	Share	Apps (voices)	Example, verbatim
Constant crashes and broken functionality Reviewers report crashes, freezing, loading or login failures, and broken features after updates that make the app unreliable or unusable.	303	16%	QuickBooks 221, FreshBooks 22, Invoice2go 15, Xero 13, Invoice Simple 13, Harvest 8, Zoho Invoice 6, Bonsai 5	> “Gets worse with every single update & is now completely unusable” receipt
Updates cause crashes and invoice failures Reviewers say recent updates have made the app slow and unstable, causing freezing or crashes and preventing invoices from loading or sending.	196	10%	QuickBooks 112, FreshBooks 34, Invoice2go 27, Invoice Simple 11, Zoho Invoice 5, Xero 5, Invoice Ninja 2	> “This app was seamless before whatever update has been given. Now takes an age to load crashes constantly while trying to send invoices.” receipt
Frequent crashes and freezing Reviewers say the app has become buggy and unstable, often freezing or crashing and preventing them from invoicing or completing other work.	181	10%	QuickBooks 154, Xero 14, FreshBooks 8, Zoho Invoice 3, Invoice Simple 1, Invoice2go 1	> “New update on this app is now unusable. unable to raise invoices on phone, as app just crashes” receipt

Account and payment problems with poor support Reviewers say blocked accounts, withheld or failed payments, and broken billing features disrupted their businesses, while customer support provided little or no effective help.	178	10%	Invoice2go 82, QuickBooks 32, Invoice Simple 23, FreshBooks 15, Xero 12, Zoho Invoice 8, Bonsai 5, Invoice Ninja 1	> “Could not process a clients payment and their service is terrible ! No phone line just messaging people that will eventually just cancel your subscription because they feel like your not worth the headache terrible.” receipt
App is limited and unreliable Reviewers find the app confusing, buggy, slow, or lacking essential functionality, making it difficult to accomplish what they need.	169	9%	QuickBooks 73, Xero 25, Invoice Simple 23, Invoice2go 20, Harvest 8, FreshBooks 8, Zoho Invoice 7, Invoice Ninja 3, Bonsai 3	> “Can’t do much on the app at all. Not very useful.” receipt
Rising fees and payment frustrations Reviewers object to rising prices, unexpected charges, high payment-collection fees, and pressure to use built-in payment processing.	155	8%	Invoice2go 43, Invoice Simple 37, Zoho Invoice 33, QuickBooks 14, Xero 12, FreshBooks 7, Harvest 5, Invoice Ninja 3, Bonsai 3	> “When I first started using this app everything was free, then they added abt a $12 charge for basically nothing, now charging to make and send invoices?” receipt
Mobile app fails at basic invoicing tasks Reviewers report that the mobile app cannot reliably perform essential tasks such as viewing, creating, editing, sending, and managing invoices or receipts.	138	7%	QuickBooks 40, Xero 30, Invoice Simple 20, FreshBooks 18, Invoice2go 14, Zoho Invoice 12, Harvest 3, Bonsai 1	> “The latest update has totally ruined it, can’t see any of my invoicing or create a new one, the buttons don’t work/show.” receipt

Receipt uploads are broken Reviewers say receipt capture and uploading have become buggy, slow, or unreliable, often preventing them from attaching receipts to expenses or transactions.	134	7%	QuickBooks 100, Xero 14, FreshBooks 9, Invoice Simple 4, Harvest 4, Bonsai 3	> “Suddenly can’t upload pics of receipts.” receipt
Rising prices and invoice limits Reviewers object to steep subscription price increases and new monthly invoice caps that force them to pay more.	125	7%	Invoice Simple 108, Invoice2go 12, FreshBooks 2, QuickBooks 2, Zoho Invoice 1	> “Have used this app for the past few years, with price increases every year and locking certain features. Now they are limiting the number of invoices you can even send per month.” receipt
Repeated price increases drive users away Reviewers say recurring price increases have made the apps too expensive, prompting some to consider switching or canceling.	116	6%	Invoice Simple 75, Xero 12, Invoice2go 11, QuickBooks 9, Harvest 8, FreshBooks 1	> “But price keep increasing every year. Last year it was $89 this year $99 and it started at $60 aprox per year. No longer rentable for me.” receipt
Mobile app is unreliable and limited Reviewers find the mobile app less dependable or capable than the web or desktop version, citing bugs, lost entries, slow syncing, extra steps, and missing features.	97	5%	Harvest 60, FreshBooks 11, Bonsai 10, Zoho Invoice 5, QuickBooks 5, Invoice Simple 3, Invoice2go 2, Invoice Ninja 1	> “App misses time entries and has lag on updates” receipt
Mileage tracking does not work Reviewers report that mileage tracking is unreliable, stops working, or fails to record trips for extended periods.	73	4%	QuickBooks 55, FreshBooks 18	> “the mileage tracking is completely broken” receipt

2. Alternatives map

For reviews of each app, how many distinct authors name another option. Matching is by name only (the model's alternative tag is NOT MEASURED). A mention is not a switch.

Reviews of	Other options named (distinct authors)
Bonsai	invoices by hand (word processor and email) 3; QuickBooks 2
FreshBooks	QuickBooks 17; invoices by hand (word processor and email) 9; a spreadsheet 1; an accountant or bookkeeper 1
Harvest	invoices by hand (word processor and email) 4; QuickBooks 2; FreshBooks 1; a self-built tool 1; an accountant or bookkeeper 1
Invoice Ninja	Harvest 1; QuickBooks 1; an accountant or bookkeeper 1
Invoice Simple	an accountant or bookkeeper 3; QuickBooks 3; invoices by hand (word processor and email) 3; a spreadsheet 2; Zoho Invoice 2; a self-built tool 1; Xero 1
Invoice2go	QuickBooks 3; invoices by hand (word processor and email) 3; a spreadsheet 2; Zoho Invoice 1
QuickBooks	invoices by hand (word processor and email) 25; an accountant or bookkeeper 18; Xero 14; a spreadsheet 1
Xero	QuickBooks 9; an accountant or bookkeeper 8; invoices by hand (word processor and email) 2
Zoho Invoice	QuickBooks 8; invoices by hand (word processor and email) 2; a spreadsheet 2; Invoice2go 1; Xero 1; Invoice Ninja 1; an accountant or bookkeeper 1

3. Riskiest assumptions

From the L1 hypothesis graph, ranked by risk priority. None has a status. Simulation cannot change these.

Node	Guess	Status	Public evidence	Test that would settle it
J2	I want to send a clear, correct invoice to the client as soon as the work is done	**unknown**	2 supporting posts, below the bar	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 2 supporting, 0 contradicting.

J3	I want to know which invoices are unpaid and how late they are	**unknown**	1 supporting post, below the bar	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 1 supporting, 0 contradicting.
C3	I can see the true unpaid balance without adding anything up myself	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.
C4	Nothing falls through the cracks because an invoice was forgotten	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.
J11	I want to see at a glance which clients owe me money and for how long	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.
P3	I still have to write and send the reminder myself, so I put it off	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.
J4	I want to remind clients to pay without damaging the relationship	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.
J5	I want to keep a record of what was agreed and billed so I can prove what is owed	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.

S2	Freelancer invoicing apps such as Bonsai or Wave	**unknown**	1 supporting post, below the bar	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 1 supporting, 0 contradicting.
J12	I want to be warned when an invoice is about to become overdue	**unknown**	no public evidence in this harvest	Supported only if at least 3 distinct authors in at least 2 customer-voice families (reviews plus a user community) describe it; contradicted on the same bar. Now: 0 supporting, 0 contradicting.

4. Strategy object

Part	Status
Target segment and Core Job	Not chosen: no segment or Job node has public support.
Value hypothesis and mechanic	Not established. The radar shows problems voiced, not a value gap (Q8 is S0).
Why we win	Not established.
Unit economics	No figure: Q4–Q6 and Q9 are S0.
Demand plan	T4 landing-page test (not run yet; B7): 3 headlines, EUR 49 vs EUR 149, about 200 founders or PMs.
Top-3 riskiest assumptions	J2: I want to send a clear, correct invoice to the client as soon as the work is done J3: I want to know which invoices are unpaid and how late they are C3: I can see the true unpaid balance without adding anything up myself Cheapest test for each: the evidence bar in section 3, then the landing-page test.

5. Quantities and decision

Q	Quantity	Grade	Why
Q1	reachable segment count	S0	no input of any family was collected in B1-B4; needs an official statistics count of the segment

Q2	job frequency	S0	no input of any family was collected in B1-B4
Q3	job budget	S0	no input of any family was collected in B1-B4
Q4	WTP	S0	simulated family dropped: cross-model rho below 0.5 and SSR gold failed; list prices of 9 alternatives exist as Wayback snapshots (B4 replay) but are not extracted into a recomputable calculation
Q5	cost-to-serve	S0	no input of any family was collected in B1-B4
Q6	CAC	S0	no input of any family was collected in B1-B4
Q7	adoption	S0	simulated family dropped: cross-model rho below 0.5 and SSR gold failed
Q8	value gap	S0	no input of any family was collected in B1-B4
Q9	retention	S0	no input of any family was collected in B1-B4

S0 means unanchored and is never cited. Monte Carlo (10,000 runs, seed 0): not run, because Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9 are below S1. EVPI: not computed.

Synthetic validation check	Result
Q1-Q4 and Q7 at least S3	fail
the rest at least S2	fail
three probabilities at least 0.6	fail
Q8 > 0 with p at least 0.8	fail
red team finds no single input that flips the result	pass

6. Red team

A different model family (GLM-5.3, thinking off) read only the report packet.

Single input that flips the conclusion: none found (The NO GO rests on structural zeros: 0 supported nodes (max observed is 2 supports vs a bar of 3 authors in 2 families), all 9 quantities S0 with no inputs collected, and Monte Carlo blocked. No single input change flips it because the evidence bar requires at least 3 distinct authors in at least 2 customer-voice families, and the harvest has only 1 non-app-store post, so even a large new app-review batch cannot satisfy the 2-family requirement; a flip requires a new evidence family (e.g. a user community harvest), which is a new collection, not a change to an existing input. The closest single candidate is J2 (2 supports, 0 contradicts): one additional distinct author from a second family would make it supported, but that alone would not flip the overall NO GO since 9 of 10 nodes would remain unsupported and all money figures would stay S0.).

Where	Claim it flagged	Why	Response
radar.groups[0..11] (all 12 groups)	12 distinct problem themes with named labels like 'Constant crashes and broken functionality', 'Updates cause crashes and invoice failures', 'Frequent crashes and freezing'	Groups 10, 11, and 2 are near-duplicates (crashes/freezing after updates) and groups 5, 0, and 7 overlap heavily (price increases); the packet presents them as separate findings without a dedup check, inflating apparent breadth of problems	Agreed. k = 12 was fixed in amendment 7 before clustering, and no merge was done afterwards. Three crash themes and three price themes overlap; read them as about 6 broader themes (crashes after updates, price and fees, account and payment support, mobile limits, receipts, mileage).
radar.groups[*].years	Trend claims implied by year breakdowns (e.g. group 10: 2023:24, 2024:22, 2025:90, 2026:167)	No denominator per year is given; rising counts may just reflect harvest volume, and '2026' reviews (167 posts) are future-dated relative to a normal calendar, suggesting a date-parsing or labeling error that is not flagged	Partly. Year counts are raw voices with no per-year denominator, so the report draws no trend from them. 2026 is not an error: the harvest ran in November 2026.
radar.by_platform	Market-level 'invoicing' problem radar	3283 of 3284 posts are App Store reviews and only 1 is Hacker News; the evidence bar requires 2 customer-voice families, so the radar is effectively a single-family (app reviews) sample and cannot support claims about the broader invoicing market	Agreed. The radar is one customer-voice family (App Store reviews, US/GB/IE). It is not a market-level claim; added to Method and limits.

radar.groups[*].apps	Cross-app problem patterns (e.g. group 0 'Rising prices and invoice limits' spread over Invoice Simple, Invoice2go, FreshBooks)	Group 0 is 108/125 Invoice Simple and group 3 is 60/97 Harvest; these are single-app complaint clusters presented as market-wide themes	Agreed. Per-app voices are in the Apps column: 'Rising prices and invoice limits' is 108 of 125 Invoice Simple, 'Mobile app is unreliable and limited' is 60 of 97 Harvest. Read single-app themes as app-specific.
simulation.conditions	Quantified switch probabilities, e.g. p_switch 0.515 for 'copy' and 0.294 for 'take_job', with 80% intervals	The packet's own validity metrics refute these numbers: price_order_preserved 8/27, repeat_mad 0.1589, sd_p_switch 0.2568, cross-model rho 0.4036 with verdict 'unusable', and T3 UNDECIDABLE; the intervals imply a precision the simulation does not have	Agreed. That is why the section is exploratory, sits last and feeds no quantity or node. The intervals are a bootstrap over cells: sampling spread only, not validity.
simulation.conditions[price10 vs price50]	A 10% price raise yields higher p_switch (0.428) than a 50% raise (0.404)	This non-monotonic result is internally inconsistent with any sane demand model and the packet's own 8/27 price-order check confirms it is noise, yet the numbers are still reported as estimates	Agreed: noise, consistent with price order preserved in only 8 of 27 cases. Shown as a failure signal, not an estimate.
status.gold	Cross-model agreement 89.7% (kappa 0.86) presented as a quality signal	Agreement is between two models (Kimi and GLM), not against ground truth; human_check is NOT DONE, so this measures model correlation, not correctness, and the label 'gold' overstates it	Agreed. The report calls these labels model-labelled (Kimi), the GLM figure cross-model agreement, and the human check NOT DONE. 'gold' is only the file name.

t1.trigger / success_criterion / alternative / aha / supports / contradicts	Precision figures (0.5, 1.0, 0.8, null) reported per label	Called counts are 6, 2, 5, 0, 1, 0; these are far too small for the stated precision values and several are marked NOT MEASURED, yet numeric precisions are still displayed as if informative	Agreed. The report shows only counts for these classes, each marked NOT MEASURED (fewer than 20 calls); no precision is claimed.
riskiest[*].priority	Priority scores (1.8, 1.5) ranking the 10 riskiest nodes	0 of 10 nodes have any supporting evidence (7 no_public_evidence, rest below bar) and the structure comes from the unvalidated L1 hypothesis graph; priorities rank unverified hypotheses	Agreed. Priority is the L1 risk ranking of unverified guesses: it orders what to test, not what is true.
alternatives (all)	Alternative-usage mapping, e.g. QuickBooks users consider 'invoices by hand' (25) and 'an accountant' (18)	These counts come from the same single-family app-review harvest with no extraction-quality check on the 'alternative' label (precision 0.8 on 5 calls); they are presented as a switching map without any status	Partly. The map does not use the model's alternative tag (NOT MEASURED); it is name matching by regex. Agreed that it has no quality check: the generic patterns (manually, accountant) also match passing mentions; added to Method and limits.
touch	'T4 landing-page test: 3 headlines, EUR 49 vs EUR 149, about 200 founders or PMs'	This is a plan, not a result; the packet gives no data from it, so listing it alongside measured sections implies a test that has not run	Agreed. It is a plan, not a result; the report now says not run (B7).

monte_carlo	Monte Carlo blocked by Q1-Q9	Not an overreach per se, but the report's NO GO / run-touch-first conclusion is presented as final while every quantity is S0 with null estimates; the conclusion is only as strong as the absence of data, which the packet does state, but any headline framing 'market too small / not validated' would overreach	Agreed. NO GO means do not build yet, because the inputs are missing; it does not mean the market is too small.

7. Simulation (exploratory, unvalidated)

Status: exploratory, unvalidated. SSR gold failed (stay 78%, switch 77%, bar 85%); cross-model agreement is unusable (lowest Spearman ρ 0.4036 on 53 cells); T3 is UNDECIDABLE (0 of 21 past price changes had enough switchers to score). This section is not evidence, does not count toward any node, and feeds no quantity.

Model: deepseek/deepseek-v4.1-flash, 3 repeats. Calibrated on 9 masked public posts: measured inputs USER and problem (precision against model labels), the 4 below-bar supporting posts, and unmeasured tags at low weight (trigger, success_criterion, alternative, aha). Interval: 80% bootstrap over agent × app × repeat cells. Repeat MAD 0.1589; price order preserved in 8/27.

If…	Mean P(switch)	80% interval	Label
Nothing on the market has changed.	0.389	0.354–0.423	**SYNTHETIC (simulation estimate)**
The current app raises its price by 10% from next month.	0.428	0.388–0.468	**SYNTHETIC (simulation estimate)**
The current app raises its price by 50% from next month.	0.404	0.367–0.44	**SYNTHETIC (simulation estimate)**
A new invoicing app launches that sends payment reminders by itself and shows every unpaid invoice in one place, at about the same price as the current app. The new app was not one of the switch destinations SSR scored (B4 offered only existing options), so switching to it is undercounted here.	0.368	0.332–0.404	**SYNTHETIC (simulation estimate)**

The current app stops helping you get paid.	0.482	0.446–0.519	**SYNTHETIC (simulation estimate)**
A service launches that chases your unpaid invoices for you, so the job is taken off your hands, for a monthly fee.	0.294	0.261–0.326	**SYNTHETIC (simulation estimate)**
A rival copies the feature you rely on most in the current app.	0.515	0.48–0.548	**SYNTHETIC (simulation estimate)**

8. Method and limits

- Producer: DeepSeek V4.1 Flash. Auditor and gold labels: Kimi K3.1 Pro. Cross-model check: GLM-5.3. Embeddings: Qwen3. All through TensorX.
- Precision on the held-out test split (against model labels): USER 75/77 (PASS); trigger 3/6 (NOT MEASURED); success_criterion 2/2 (NOT MEASURED); alternative 4/5 (NOT MEASURED); problem 44/44 (PASS); aha 0/0 (NOT MEASURED); supports 1/1 (NOT MEASURED); contradicts 0/0 (NOT MEASURED). Fewer than 20 calls means NOT MEASURED. Human label check: NOT DONE.
- T2 failed: supports plus contradicts had fewer than 20 test calls, so no node can be supported or contradicted. The product narrowed to this Radar.
- Pool limits: App Store reviews from 9 allow-listed invoicing apps (US, GB, IE); the two unrelated "Wave" listings are excluded; there is no Reddit collector; star ratings sit in the review text but are not analysed; only 1 of the 16 HN USER posts is in this harvest. So the radar is one customer-voice family, not a market-level picture.
- Themes: k = 12 was fixed before clustering and nothing was merged afterwards, so some themes overlap (three about crashes, three about price). Some themes come mostly from one app; the Apps column shows the split. Year counts have no per-year denominator and are not trends.
- Alternatives: app names are matched by regex; the generic options (by hand / manually, accountant, spreadsheet) also match passing mentions such as "I have to enter it manually", so those counts are upper bounds.
- Voices are counted once per salted author hash. No names, emails or private data are stored.